1.

```
Booting from Hard Disk..xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inodes8
init: starting sh
$ pttest
Page table before sbrk:
Page table for pid 3:
VA 0x0 -> PA 0xdee2000 | P=1 W=1 U=1
VA 0x2000 -> PA 0xdedf000 | P=1 W=1 U=1
Total valid user pages: 2
User page-table pages: 1

Page table after sbrk(4 * 4096):
Page table for pid 3:
VA 0x0 -> PA 0xdee2000 | P=1 W=1 U=1
VA 0x2000 -> PA 0xdedf000 | P=1 W=1 U=1
VA 0x3000 -> PA 0xdfbc000 | P=1 W=1 U=1
VA 0x4000 -> PA 0xdf76000 | P=1 W=1 U=1
VA 0x5000 -> PA 0xdfbf000 | P=1 W=1 U=1
VA 0x6000 -> PA 0xdfc0000 | P=1 W=1 U=1
Total valid user pages: 6
User page-table pages: 1
$
```

2, 3.

```
Booting from Hard Disk..xv6...
cpu0: starting 0
sb: size 1000 nblocks 941 ninodes 200 nlog 30 logstart 2 inode8
init: starting sh
$ faulttest
pagereplReserved 32 pages of virtual memory
Pages touched: 5
Additional faults from page writes: 5
PASS: lazy allocation test
$
Reference string: 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

Frames: 3
FIFO page faults: 15
FIFO hits: 5, hit ratio: 25%
LRU page faults: 12
LRU hits: 8, hit ratio: 40%

Frames: 4
FIFO page faults: 10
FIFO hits: 10, hit ratio: 50%
LRU page faults: 8
LRU hits: 12, hit ratio: 60%

Frames: 5
FIFO page faults: 9
FIFO hits: 11, hit ratio: 55%
LRU page faults: 7
LRU hits: 13, hit ratio: 65%
$
```